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The neural and physiological substrates of real-world attention change across development.

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Abstract

The ability to allocate and maintain visual attention enables us to adaptively regulate perception and action, guiding strategic behaviour within complex, dynamic environments. This capacity to regulate attention develops rapidly over the early years of life, and underpins all subsequent cognitive development and learning. From screen-based experiments we know something about how attention control is instantiated in the developing brain, but we currently understand little about the development of the capacity for attention control within complex, dynamic, real-world settings. To address this, we recorded brain activity, autonomic arousal and spontaneous attention patterns in N=58 5- and 10-month-old infants during free play. We used time series analyses to examine whether changes in autonomic arousal and brain activity anticipate attention changes or follow on from them. Early in infancy, slow-varying fluctuations in autonomic arousal forward-predicted attentional behaviours, but cortical activity did not. By later infancy, fluctuations in fronto-central theta power associated with changes in infants' attentiveness and predicted the length of infants' attention durations. But crucially, changes in cortical power followed, rather than preceded, infants' attention shifts, suggesting that processes after an attention shift determine how long that episode will last. We also found that changes in fronto-central theta power modulated changes in arousal at 10 but not 5 months. Collectively, our results suggest that the modulation of real-world attention involves both arousal-based and cortical processes but point to an important developmental transition. As development progresses, attention control systems become dynamically integrated and cortical processes gain greater control over modulating both arousal and attention in naturalistic real-world settings.

eLife assessment

This **important** study explores infants' attention patterns in real-world settings using advanced protocols and cutting-edge methods. The presented evidence for the role of EEG theta power in infants' attention is **solid**. The study will be of interest to researchers working on the development and control of attention.

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1. Introduction

The ability to allocate and maintain visual attention enables the flexible regulation of perception and action that is characteristic of strategic behaviour (1,2). The capacity to pay attention develops rapidly over the early years of life (3), and individual differences in early attention predict long-term cognitive and clinical outcomes (4,5). Recent new methodological advances such as naturalistic neuroimaging are allowing us to build on previous research using lab-based behavioural experiments and animal studies.

The development of attention is traditionally conceptualised as the product of interactions among different systems at different levels of maturity (1,6,7). Traditionally, the earliest subcomponent of attention to develop is thought to be the arousal/ alertness subcomponent, mediated via brainstem reticular activating systems centred on the locus coeruleus (LC) and instantiated primarily via norepinephrine neurotransmitter systems (6). In young infants, alertness is more readily initiated by exogenous events (8); over time, infants gain the ability to both attain and maintain an alert state even in absence of external stimulation. Areas around the brainstem (including the LC) are thought to be some of the earliest to become functionally mature (9,10). Consequently, the relative influence of this subcomponent of attention is thought to be strongest during early development (6).

Behaviourally, the arousal/ alertness subcomponent of attention is thought to reflect a state of anticipatory readiness, or alertness for stimulus input (6). Arousal is generally measured indirectly, via proxy measures of autonomic nervous system activity such as heart rate (11). Heart rate has been extensively studied in the context of infant attention (3,12,13). During anticipatory readiness, we know that reorientations of visual attention take place periodically, clustered around a preferred modal reorientation rate (14–18). This may reflect rhythmic activity in the central nervous system (19).

With time, it is thought that looking behaviours become increasingly modulated by higher-level executive processes that reflect the infant's internal states, motivation, comprehension, and goals (2,20,21). Behaviourally, this increase in endogenous or internally directed attention has been shown as: a developmental increase in the degree to which attentional engagement is accompanied by decreases in distractibility (22,23); an increase in selective attention as measured indirectly, using the blink reflex (24); and differences in the trajectory of how attention durations to simple vs complex stimuli change over developmental time (25).

Other research that used experimenter-controlled, screen-based tasks to examine neural correlates of attention has examined changes in the power spectral density (PSD) of EEG oscillations, in particular infants' theta (3–6Hz) rhythm, which increases during active, anticipatory, and exploratory behaviour (26–31). Together, these studies suggest that the expression of theta during attention-eliciting episodes could signify the engagement of neural networks related to executive attention (28,29,31). Similarly, other studies have reported decreases in alpha band activity under conditions of increased attention (32,33). Both theta and alpha effects are now widely known in the literature as "theta synchronization" and "alpha desynchronization" (29).

How children allocate their attention in experimenter-controlled, screen-based lab tasks differs, however, from actual real-world attention in several ways (34–36). For example, the real-world is interactive and manipulable, and so how we interact with the world determines what information we, in turn, receive from it: experiences generate behaviours (37). While lab-based studies can be made interactive (e.g., 38,39), how infants actively and freely initiate and self-structure their attention remains unexplored (40).

The present study aims to examine developmental changes in the relationship between autonomic arousal, cortical activity, and attention in real-world settings. To do this, we first explored how naturalistic attention patterns (measured via looking durations to play objects) from a solo play interaction change between 5 and 10 months. Then, we explored temporal relations between changes in infant arousal (measured via heart rate) and attention episodes in typical 5- and 10-month-olds infants. Finally, we investigated changes in EEG theta power relative to attention episodes, and changes in EEG theta relative to arousal (see **Figure 1**). As attentional systems mature and brain regions become increasingly specialised (41,42), we expected to see both a developmental increase in attention towards play objects and a developmental shift in the way different mechanisms (i.e., arousal/ alertness vs. executive attention subsystems) drive attention. To measure attention we used looking time, an approach that is known to have several limitations (12,43,44). For example, we cannot differentiate based on looking time alone whether overt and covert attention are coupled or decoupled (see 3.1).

Our first set of analyses examined attentional inertia (the phenomenon that, as individuals become progressively more engaged with an object, their attention progressively increases) as a measure of internally driven attentional engagement (3,45,46). We tested whether attentional inertia influenced attentional behaviours more strongly at 10 months compared to 5 months. To do so, we calculated both the Autocorrelation Function (ACF) and the survival probability of spontaneously occurring attention episodes during play (analysis 1). The ACF allowed us to quantify the rate of change of spontaneous attention durations. A faster rate of change would indicate lower attentional inertia. The survival probability, on the other hand, allowed us to quantify the probability between looking (i.e., paying attention) and looking away. A slower decrease in the probability of an attention episode surviving would indicate increased attention engagement and decreased distractibility by other stimuli. We hypothesised that, as slow-varying fluctuations in endogenous interest or engagement start to influence looking behaviour more strongly over time, 10-month-old infants would show increased endogenous attention control indexed by a slower rate of change of attentiveness and slower decreases in the survival probability. We also predicted that we would be able to identify periodic attentional reorientations during early as well as later development (17,47); later in development, however, we predicted that infants would be more likely to extend visual fixations beyond their modal periodic reorientation rate, possibly indicating a greater or more efficient integration of attention and gaze shifting (48), and that attention duration episodes would be longer overall.

Next, in order to assess the link between lower-level mechanisms of autonomic arousal and attention, we calculated cross-correlations between autonomic arousal (indexed via heart rate) and attention episodes across the entire play session for both 5- and 10-months olds. This allowed us to examine whether arousal changes tend to forward-predict changes in attention, or vice versa (analysis 2). Based on previous research (49), we hypothesized that periods of elevated autonomic arousal would associate with, and forward-predict, shorter attention episodes. We also predicted that such relationship would weaken with time due to the maturation of cortical attentional systems.

We used a similar approach to examine developmental changes in the relationship between neural markers of executive attention and real-world attention behaviours. We were interested to examine whether neural changes (indexed by theta power) anticipate subsequent attentional behaviour shifts (48); or, whether neural processes after the attention shift relate to increases in infants' attention engagement. To test this, we conducted three analyses. First, we analysed neural activity across a range of time windows both before and after the onsets of new attention episodes and performed linear mixed effect models to examine how neural activity before and after attention onset associated with the subsequent durations of those episodes (analysis 3). Second, we examined changes in neural activity during individual attention episodes (analysis 4). Finally, we used cross-correlations to examine whether, across the entire dataset, neural markers tend to forward-predict changes in attention, or vice versa (analysis 5). We predicted that the associations

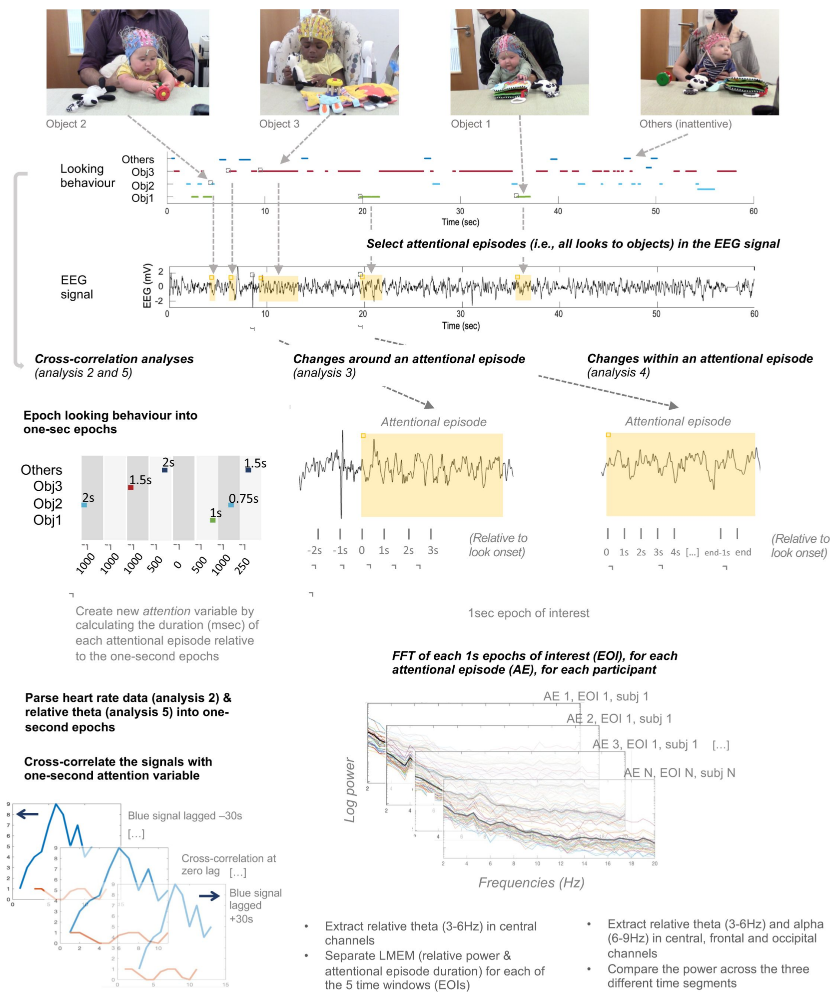

Figure 1.

Experimental set up and schematic illustration of the procedure followed for analysis 2 to 5.

Top figure shows the experimental set up for solo play. Below, on the left, is shown the procedure followed to parse the looking behaviour and create the variable "attention", and further cross-correlation analyses (analysis 2 and 5). On the right, instead, is shown the steps followed to identify attentional episodes in the EEG signal and further EEG analyses (analysis 3 and 4).

between neural markers of executive attention and real-world attention behaviours would become stronger with increasing age (i.e., theta activity would show a stronger predictive relation with infants' attentional behaviours at 10 months, as evidence of increased modulatory power from the executive attention system on infants' attention).

Finally, we examined whether there were any interdependencies between autonomic arousal and theta activity. We had no predictions for how this relationship would change over time.

2. Results

2.1 Analysis 1: Developmental changes in attention

Our first set of analyses examined attentional inertia as a measure of internally driven attentional engagement. Attention inertia is the phenomenon that, as individuals become progressively more engaged with an object, their attention progressively increases. We tested whether attention inertia is stronger at 10 months compared to 5 months. To do so, we calculated the ACF and the survival probability of spontaneously occurring attention episodes during play to quantify the rate of change of spontaneous attention durations and the probability between looking (i.e., paying attention) and looking away, respectively.

Initially, we conducted four descriptive analyses to test how attention and inattention durations and reorientations change over both the course of the solo play interactions and developmental time. First, we tested how many times per minute 5- and 10-months-old infants redirected their attention from one object to the other. We found that, on average, 5-month-old infants performed significantly more both attentive ($t(10)= 4.346$, $p=0.001$) and inattentive ($t(10)= 4.202$, $p=0.002$) reorientations during the solo play interaction than 10-months-old infants (**Figure 2A**). When we looked at how attention reorientations changed during the course of the solo play episode, we found that 5-month-old infants performed consistently more looks than 10-month-olds throughout the interaction even though the number of looks per minute decreased over the course of the interaction for both age groups (**Figure 2B**, and Figure S1A and B).

Second, we investigated the average duration that 5- and 10-month-old infants spent in attentive and in inattentive states during the solo play interaction and minute by minute (**Figure 2C and D** respectively). In general, infants' attention durations toward play objects at 10 months were longer ($t(10)= -2.787$, $p=0.019$). At 5 months, moments of inattention were longer than moments spent looking towards the object ($t(10) = -3.749$, $p=0.003$). Overall, at 10 months, but not 5 months, infants spent more time in attentive compared to inattentive states ($t(58)= 10$, $p<0.001$) (**Figure 2E**). We then calculated a best fit line, individual by individual, to look at how average attention duration changed within the session (see Figure S1C and D). We found no significant differences in the way attention duration changed during the interaction between the two age groups (Figure S1C and D).

Third, we explored the distribution of looks towards the objects (**Figure 2F**). At both ages, attention durations shorter than or equal to 5 seconds follow a positively skewed lognormal distribution, with modal attention durations in the 0.5 – 0.6 second range. Modal attention durations were significantly lower at 5 than at 10 months ($t(58)= 2.211$, $p=0.03$). Finally, the right plot of **Figure 2F** shows extended attention episodes. There was an increasing amount of such looks with increasing age.

Following the descriptive statistics, we calculated both the ACF and the survival probability of the looking behaviour (**Figure 3**). First, we used time-series analyses to examine the rate of change of attention durations, relative to itself. We calculated the ACF of the attention durations at both time points (more details in 4.3.5.2). The ACF indexes the cross-correlation of a measure with itself

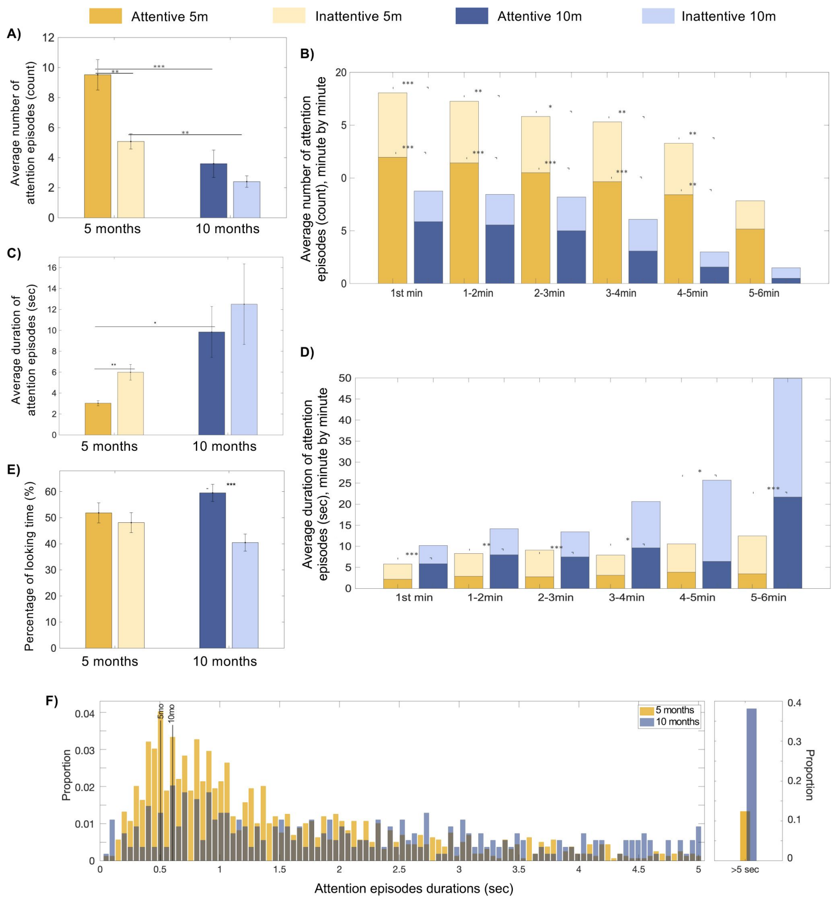

Figure 2.

Descriptive analyses on infant attentional behaviour during the solo play interaction.

(**A**) Average number of attentive and inattentive looks per minute at 5 months (left) and 10 months (right), (**B**) Average number of attentive and inattentive looks minute by minute. Asterisks show the significance values of comparisons examining how average number of looks per minute of the interaction differed as a function of age. (**C**) Average duration spent in one of the two possible attentional states: attentive and inattentive, and (**D**) minute by minute. Again, asterisks show the significance values of comparisons examining differences as a function of age. (**E**) Percentage of time infants spent in attentive vs. inattentive states, during the whole interaction. (**F**) Histogram showing the distribution of the proportion of all the looks that lasted less than or equal to 5 seconds (right) and more than 5 seconds (left) at 5 months (yellow) and 10 months (blue). Continuous black line indicates the mode of each distribution. Significance is indicated with asterisks where * = p<0.05, ** = p<0.01, and *** = p<0.001. Error bars represent SEMs.

at different lag-intervals in time (46). ACF values were obtained from 0 to 10 seconds lag, in steps of 500 milliseconds. As shown in **Figure 3A**, the ACF of the time series looking behaviour fell off more sharply at 5 months than at 10 months. The ACF values were compared across ages using independent sample t-tests. From lag +500 milliseconds to 10 seconds, 10-months-old infants showed significantly higher correlation values than 5- months-old infants.

Second, we performed a survival analysis by calculating the survival probability function of the looking behaviour towards the objects at both time points. The survival probability function is the probability that an attention episode survives longer than a certain time. As shown in **Figure 3B**, the survival probability of a look decreased abruptly at the beginning, for the very short looks, and flattened as looks got longer. The differences in the speed at which the survival probability decreased can be seen more clearly by calculating the derivative of the survival probability (**Figure 3C**). To compare survival among the two groups, we performed the log rank test using the Matlab function 'Logrank' (50). The results for the log-rank test rejected the null hypothesis ($p<0.001$) indicating that the survival curves for looking behaviour at 5 months and 10 months were significantly different. Notably, the likelihood of a look ending is more tightly clustered around the modal value of 0.5 seconds at 5 months.

Overall, our results showed that older infants demonstrated to have a slower-changing profile of attention with longer attention episodes overall (**Figure 2A and B**, **Figure 3A**). At both ages, there was evidence for a preferred modal reorientation rate in the 0.5–0.6 second range, which was slightly faster at 5 months than 10 months (**Figure 2F**). Attention durations were more tightly clustered around the modal value at 5 months. At 10 months, attention episodes were more likely to be extended beyond the preferred modal reorientation rate than at 5 months (**Figure 3B and C**).

2.2 Analysis 2. Auto- and cross-correlation analyses between infant autonomic arousal and attention

In this section we investigated the relationship between changes in infant autonomic arousal (indexed by heart rate activity) and their associations with moment-to-moment changes in attention (indexed as a continuum of looking durations to play objects vs. elsewhere).

Figure 4 (A and C) shows the results of the autocorrelation analyses for autonomic arousal at 5 and 10 months of age respectively. Significant autocorrelations were observed at relatively short lags around t=0 (from -4 to +4s) at both ages. **Figure 4 (B and D)** shows the results of the cross-correlation analysis between autonomic arousal and attention at 5 and 10 months of age respectively. The negative values indicate that, at 5 months, lower heart rate forward-predicted increased looking durations from lags between -9 to -2 seconds (i.e., lower heart rate at time t significantly associated with increased attention at time t+9 seconds). The same pattern was present but not significant at 10 months. The asymmetry of this cluster around the lag t=0 indicates that changes in heart rate tended to forward-predict changes in attention more than vice versa.

2.3 Analysis 3. Calculation of neural power changes around an attention episode

We used linear mixed effects models to examine the associations between the length of each attention episode (i.e., looking duration to any of the play objects) and relative theta power at different time windows relative to the onset of that attention episode (see **Figure 5**). At 10 months, relative theta power in the time window of 0 to +1000msec and +1000 to +2000msec after onset of a new attention episode predicted the subsequent duration of that attention episode. At 5

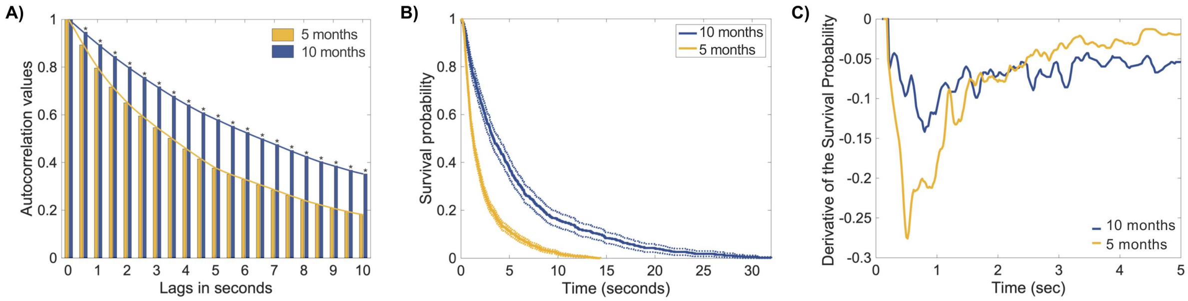

Figure 3.

ACF and survival probability analyses of the looking behaviour.

(**A**) Autocorrelation of the time series looking behaviour at 5 months (in yellow) and 10 months (in blue). (**B**) Survival analysis. Survival probability function for looking behaviour toward object toys. The survival function is the probability that a look will survive a given time. Yellow line shows data from 5-months-old infants with confidence bounds (dotted yellow line) and blue line shows data form 10-months-old infants with confidence bounds (dotted blue line). (**C**) Derivative of the Survival Probability at 5 months (yellow) and 10 months (blue).

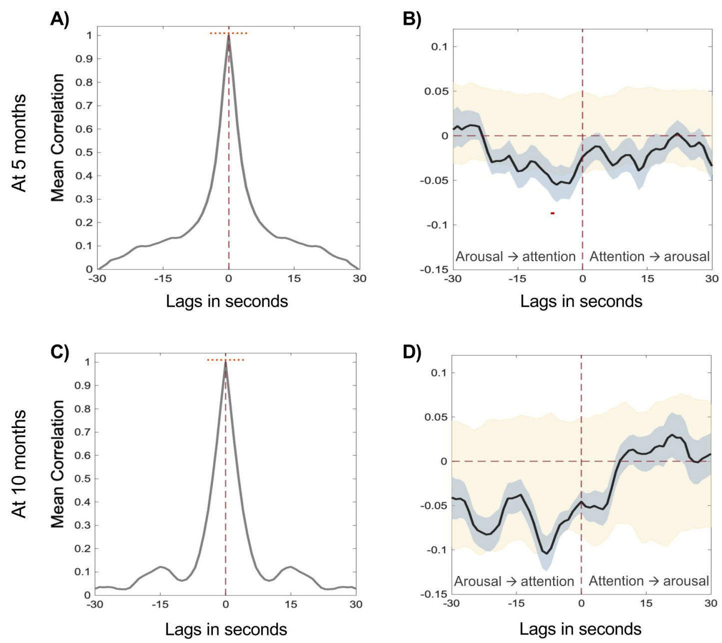

Figure 4.

Relationship between infant autonomic arousal and attention.

Autocorrelation results for infant autonomic arousal at 5 months (**A**) and 10 months (**C**). Significant clusters are indicated by red dots. Cross correlation between infant autonomic arousal and attention at 5 months (**B**) and at 10 months (**D**). Infant autonomic arousal forward-predicting infant attention on the negative lags, infant attention forwardpredicting infant autonomic arousal on the positive lags. Black lines show the cross-correlation values, shaded grey areas indicate the SEM. Shaded yellow areas show confidence intervals from the permuted data. Significant time lags identified by the cluster-based permutation analyses are shown by a thick red line.

months, the same relationships were not significant. We found no evidence of neural activity before the start of an attention episode forward-predicting the length of that attention episode at any time point (**Figure 5**).

The final number of accepted trials (i.e., attention episodes) in the analyses varied across the three timewindows immediately after the onset of each look. More trials were obtained for the first window (total number of looks at 5 months was 790, and 411 at 10 months) than the second (total number of looks at 5 months was 473, and 336 at 10 months) and the third (total number of looks at 5 months was 301, and 277 at 10 months). All three conditions ended up with enough number of clean trials that was greater than the recommended number of trials in the infant EEG literature (51–53). Thus, the differences between the number of trials for each time window are not expected to contribute to the results described above. However, we repeated this analysis by matching the number of attention episodes at 5 months to the ones analysed at 12 months. We found no differences in the results (see Figure S2).

2.4 Analysis 4. Calculation of neural power changes within an attention episode

In addition to the previous analyses, which examined the associations between the length of each attention episode and relative theta power at different time windows relative to the onset of that attention episode, we also wished to examine whether power at the theta and alpha band changed significantly during an attentional look (i.e., any look at a play object) (**Figure 6**). Relative theta was analysed as a function of these three factors: time within an attentional episode, brain areas and age with a 3-way ANOVA (**Figure 6**). There was no statistically significant interaction between the three factors. However, the analysis revealed two simple two-way interactions: one between time within an attention episode and age, $F(2) = 5.58$, $p < .005$ and the other between channel cluster and age $F(2) = 11.98$, $p < .001$. Next, we performed a multiple comparison test to find out which groups of factors were significantly different. Results are shown in table S1-S3. A follow up analysis showed a significant effect of "time within an attentional episode": 10-months-old infants had greater theta during the third-to-fourth second into the look (middle) than the first second (start) in both the central and the frontal poles. These effects were not present in 5-months-old infants. Similarly, relative alpha was also analysed as a function of these three factors: time within and attentional, brain areas and age with a 3-way ANOVA. We found no statistically significant interactions. Results are shown in Figure S3.

Again, the final number of accepted trials (i.e., attention episodes) in the analyses varied across the three time-windows into each look. More trials were obtained for the first-second window (total number of looks at 5 months was 791, and 415 at 10 months) than the third-to-fourth second (total number of looks was 172 at 5 months, and 194 at 10 months) and the last second before look termination (total number of looks was 476 at 5 months, and 337 at 10 months).

2.5 Analysis 5. Auto- and cross-correlation analyses between infant theta activity and attention

In this section we investigated the relationship between dynamic changes in infant endogenous brain activity and their associations with moment-to-moment changes in attention (measured as a continuum of looking time durations to play objects vs. elsewhere). **Figure 7A and D** shows the results of the autocorrelation analyses for infant theta activity. **Figure 7B and E** shows the results for the cross-correlation analyses between infant theta activity and infant attention. Cluster-based permutation analysis revealed a significant positive association between the two variables (marked with a red line) at 10 months around time lag=0. More specifically, increases in infant theta activity at 10 months were significantly correlated with fluctuations in infant attention (**Figure 7E**). No associations were found between theta activity and infant attention at 5 months of age. Interpreting the exact time intervals over which a cross-correlation is significant

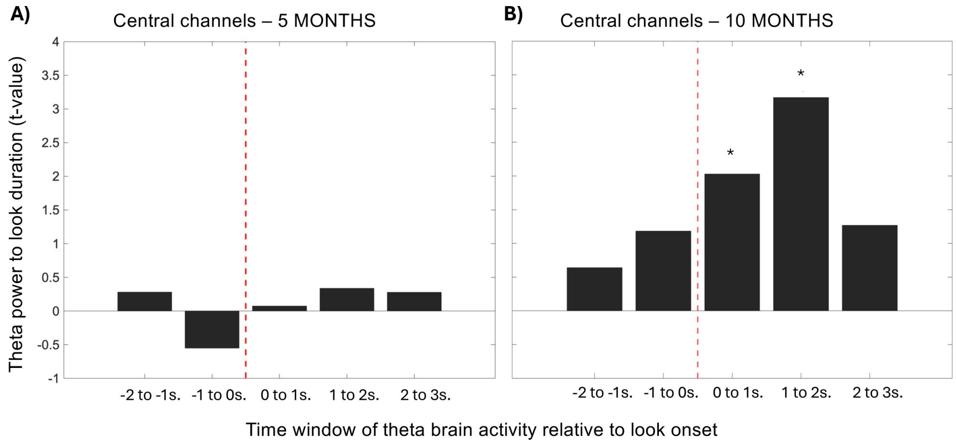

Figure 5.

Calculation of theta power changes around an attention episode.

Results of the linear mixed effects models conducted to examine whether individual looks accompanied by higher theta power are longer lasting. For each look, we calculated the association between the total duration of the look and relative theta power during five time-windows (-2000msec to -1000msec and -1000msec to 0 prior to the look, and 0 to 1000msec, 1000 to 2000msec and 2000 to 3000msec before the look), using a series of separate linear mixed effects models. (**A**) Shows results at 5 months where the y-axis is the t value, and (**B**) shows the results at 10 months. Asterisks (*) indicate p values < .05. Central channels include: 'FC1', 'FC2', 'C3', 'Cz', 'C4', 'CP1' and 'CP2'.

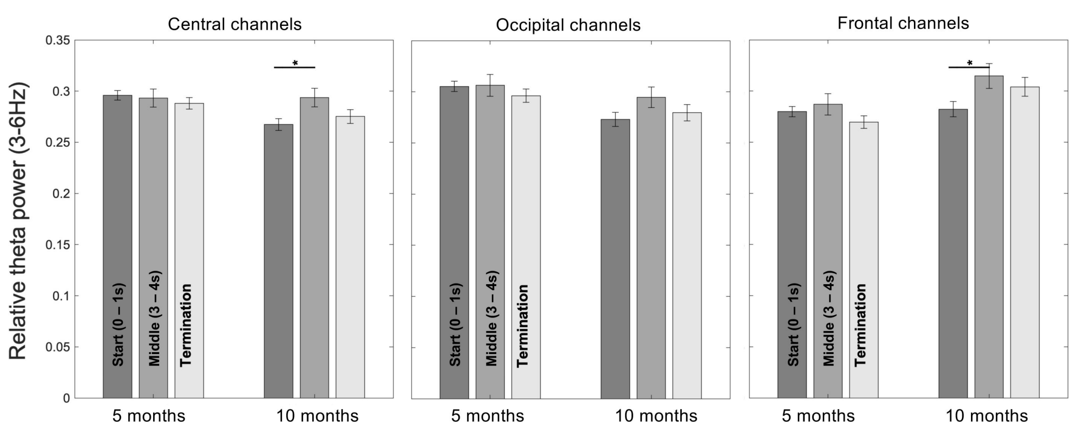

Figure 6.

Calculation of theta power changes within an attention episode.

Bar plots for the average relative theta power throughout a look, at both time points (5 and 10 months) and at different brain networks (central, occipital, and frontal). Asterisks (*) indicate p<0.05. Error bars represent SEMs.

is challenging due to the autocorrelation in the data (54,55), but the fact that the significance window is asymmetric around time 0 indicates a temporally specific relationship between infant attention and theta power, such that attention forward-predicts theta power more than vice versa.

Finally, to test whether there were any interdependencies between autonomic arousal and brain activity, we performed a cross-correlation analysis between these two variables. We found a significant cluster at 10 months (**Figure 7F**) but not at 5 months (**Figure 7C**). The asymmetry of this cluster around t=0 indicated that changes in brain activity tended to precede changes in autonomic arousal more than vice versa.

3. Discussion

We examined developmental changes in the physiological and neural correlates of real-world attention patterns during early development. To do so, we measured attention durations (to an accuracy of 50Hz), along with cortical neural activity (EEG) and autonomic arousal (via ECG) from typical 5- and 10-month-old infants playing alone while seated at a tabletop with 3 toys. This age range is a key period for early cognitive development, as differential patterns of brain development (10) drive a transition from primarily subcortical to cortical control (56), and early-emerging atypicalities can have life-long consequences (57,58). However, many of the mechanisms that drive early development remain unclear.

From Analysis 1 we found that infants at both ages showed a preferred rate of reorientation (i.e. their visual attention took place periodically). The modal durations of attention episodes towards different play objects were in the 0.5–0.6 second range at both ages but were lower at 5 months (**Figure 2F, 2C**). This contrasts with analyses of micro-level fixation durations (time intervals between individual refoveating eye movements), which decreases from early infancy (~0.5 secs) through to later infancy (~0.4 secs) through to adulthood (~0.3 secs) (15,18). Research with adults suggests that the minimum time necessary to plan and executive a saccade is ~80msecs in adults (16). Although the equivalent figure is not known in infancy, the fact that modal attention durations towards objects were shorter at 5 months than 10 months, whereas fixation durations decrease with age, makes it likely that the figures we observed do not simply indicate that infants were reorienting at the fastest speed possible, but rather were reorienting according to a preferred modal reorientation rate (15).

The survival analysis showed that, at both ages, looks were fragile early in their existence and most likely to terminate in the <1 second range (45) but the speed at which the survival probability curve decreased was faster at 5 months, meaning that the probability of a look lasting longer than time t was lower at 5 months. Richards and colleagues have found similar relationships in infants in both lab-based and naturalistic settings (45). Overall, attention durations were shorter at 5 months; this faster-changing pattern of attention to the object was also reflected in the ACF of their looking behaviour, which decreased significantly faster, showing lower overall self-similarity. Collectively, these data fit well with what we know about the development of attention. With time, we seem to observe a higher-level control of attention that allowed older infants to prioritize the task at hand – learning about/ exploring the toys – as well as to inhibit the tendency to shift attention away from an interesting task (6,20,25). Alternatively, longer attention episodes might arise because children physically manipulate objects, bringing objects closer to themselves which makes them more exogenously salient (37,59). In this case, then the infant's increased looking behaviour would be the result of increased exogenous attentional capture rather than an increase in endogenous attention control (46).

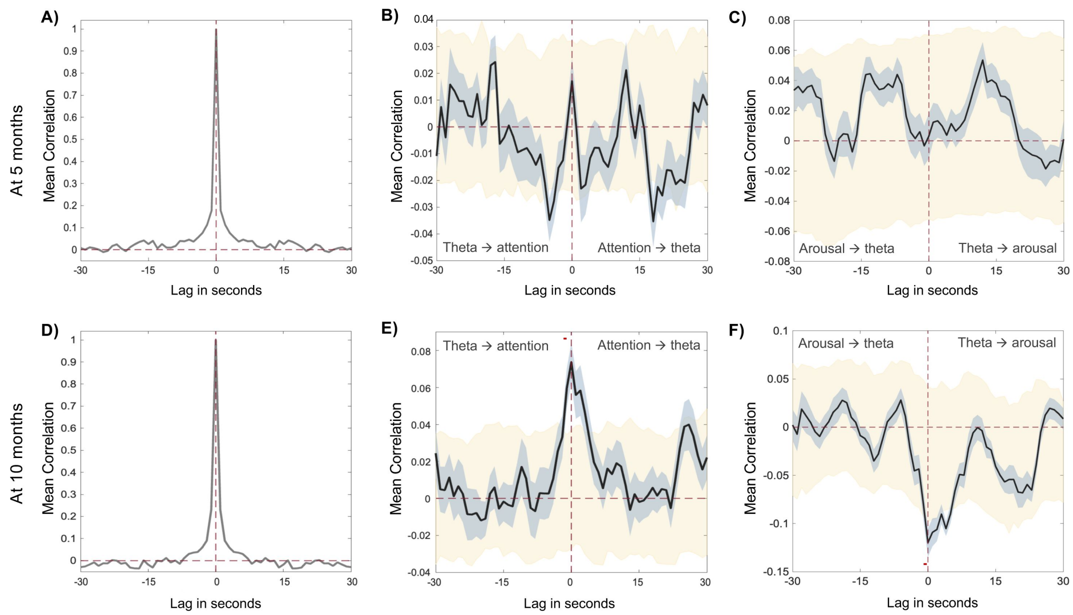

Figure 7.

Relationship between infant relative theta activity, infant attention, and infant autonomic activity.

Autocorrelation for infant theta activity at 5 months (**A**) and at 10 months (**D**). Significant clusters are indicated by red dots. Cross-correlation results between infant theta activity and infant attention at 5 months (**B**) and at 10 months (**E**). Infant theta activity forward-predicting infant attention on the negative lags, infant attention forward-predicting infant theta activity on the positive lags. Cross-correlation results between infant autonomic activity (indexed by heart rate activity) and relative theta power at 5 months (**C**) and at 10 months (**F**). Infant autonomic activity forward-predicting infant theta activity on the negative lags, infant theta activity forward-predicting infant autonomic activity on the positive lags. Black lines show the Spearman correlation at each time lag, shaded grey areas indicate the SEM. Shaded yellow areas show confidence intervals from the permuted data. Significant time lags identified by the cluster-based permutation analyses are shown by a thick red line.

Analysis 2 examined how dynamic fluctuations in autonomic arousal relate to moment-to-moment changes in attention. Consistent with previous work (13,49,60), the average concurrent correlation between autonomic arousal and attention was negative at both age points, indicating that lower arousal was associated with increased likelihood of attention. Such links have been considered within the developmental attention regulation literature, where increases in arousal are thought to lead to distraction or difficulties focusing attention, and vice versa (61). We also found that arousal levels were significantly forward predictive of attention at 5 months but not at 10 months (**Figure 4B and D**). Theoretically, if attentional episodes drive decelerations in the heart rate (12), and older infants show longer attentional episodes on average, then one could hypothesise that older infants ought also to show a more stable pattern (i.e., higher autocorrelations) in their heart rate fluctuations than younger infants. However, this was not what we observed (**Figure 4A and C**). Overall, the much shorter attention durations observed in this setting, compared with screen-based TV viewing (12), means that heart rate decelerations relative to individual attention episodes were observed infrequently in our data. However, our data did suggest, consistent with previous research, that at 5 months, changes in autonomic arousal forward-predict subsequent changes in attention.

In Analysis 3 we examined the associations between attention episode durations and theta power either before, or after, onset of that attention episode. At 10 but not 5 months, increased theta during the period immediately after the onset of a new attention episode (0–2000msec) forward-predicted the subsequent length of that attention episode (**Figure 5**). At neither age, however, did cortical neural activity before the onset of an attention episode forward-predicted attention durations.

In Analysis 4 we examined whether cortical neural activity changed significantly during an attention episode. Consistent with previous research (29), theta power in central and frontal electrodes increased significantly during an attention episode at 10 months, but not at 5 months (**Figure 6**). Contrary to our expectations, we did not find a link between attenuated alpha during an attention episode at any age (see Figure S3).

In Analysis 5 we investigated the relationship between dynamic changes in infants' theta activity and moment-to-moment changes in attention. We identified a significant positive association between infant theta activity and infant attention at 10 months but not at 5 months (**Figure 7B and E**). Interpreting the exact time intervals over which a cross-correlation is significant is challenging (54,55), but the asymmetry of the cluster around time 0 indicates that attention forward-predicted theta power more than vice versa, consistent with the findings from Analysis 3.

These findings are consistent with previous research suggesting that, by 10 months, but not during early infancy, theta oscillations increase during sustained attention and encoding (29,41,62) and associate with longer attentional periods (63). Importantly, though, we found no evidence that endogenous neural markers before the onset of an attention episode forward-predict the length of an attentional episode at either age. Instead, what we found suggests that neural activity shortly after the onset of an attention episode forward-predicts the length of that episode. One possible interpretation of this is that neural activity associates with the maintenance more than the initiation of attentional behaviours (64).

Finally, we examined the relationship between theta power and autonomic arousal (**Figure 7C and F**). A cross-correlation analysis found a negative forward-predictive relationship between the two, such that increases in theta forward-predicted decreases in autonomic arousal at 10 months, but not at 5 months. This suggests that changes in the brain activity could be modulating subcortical changes (i.e., changes in the heart rate) and may thus be able to initiate or maintain states of arousal that are common to vigilant or sustained attentional states (6,12). Overall, it appears that, by 10 months, the different substrates of attention are more inter-linked, and stronger associations are emerging between behaviour, cortical activity, and autonomic arousal (65).

In summary, our results suggest that, earlier in development, attentional episodes are more influenced by lower-order endogenous factors such as a general arousal system (3,12) - that might reflect a stronger influence of subcortical structures over the modulation of attention - and show a preferred modal reorientation timer - which characterises infants' attention shifting more strongly. Such factors would also be present at older ages; however, their association with attention would weaken over developmental time due to the maturation in cortical attentional areas thought to take place throughout the first year of life. Later in infancy, cortical neural activity reliably changes during attention episodes, but does not forward-predict attention at either age; rather, it seems that neural changes associate with the maintenance more than the initiation of attentional behaviours. Overall, the modulation of attention seems to involve both arousalbased and cortical processes. With developmental time, however, the latter increases its control over the modulation of both (i.e., overt attentional behaviours and arousal), resulting in a more inter-linked system where associations between attentional systems are stronger. Theoretically, this is consistent with what we know about the development of executive attention from experimental and neuroanatomical studies.

3.1 Limitations and strengths

Our findings should be interpreted with consideration to a number of limitations of the study. First, our events of interest are intrinsically linked with one of the biggest EEG artefacts (i.e., eye movements), and so it is possible that residual artifact in the EEG signal may have contaminated our data. However, our data were processed using algorithms specially designed to clean naturalistic EEG data (66,67), and previous analyses suggest that the electrode locations and frequency bands that we examined should be least affected by artifact, compared with more anterior locations and higher and lower frequencies (68). Additionally, our analyses were carefully designed to preclude this potential confound. First, our analyses compare events that we know share the same level of artefact/ noise (i.e., saccades at 5 months old contribute to comparable noise levels than at 12 months old (69); second, analysis 3 and 4 are time-locked to a saccade to eliminate the possibility that saccadic frequency may have influenced our results; and third, other research (70) suggest that artifact associated with saccades disappears within ~300msecs, whereas the associations between theta and look duration lasts much longer than this, up to ~6 seconds. For all this, we consider that the possibility that our results may have been caused by infants' saccades is unlikely.

Second, while the study of theta and alpha activity can offer insights into infants' intrinsically guided attention beyond its behavioural manifestations (26), attributing functional significances to particular frequency bands can be risky, especially when the ways in which they are controlled and the extent to which they interact across attention-related brain networks, still remain largely unknown (71). Similarly, the fact that certain frequency bands seem to covary with attention does not exclude the possibility of their correlation with other processes as well.

Third, the use of different EEG systems (32- vs. 64-channel BioSemi gel-based ActiveTwo) and age groups might have contributed to the differences we observed over time. However, we compared the EEG signal quality between groups and found no significant differences (Table S5–6, Figure S4).

Fourth, we used a different set of toys at the two ages (see Figure S5). Consequently, this introduced a new source of variation (i.e., toy characteristics) that might have contributed to any of the observed differences (20). However, we chose to present developmentally appropriate stimuli at the two ages to ensure that the cognitive demands were similar at the two ages. Thus, while still possible, it is unlikely that the developmental differences observed in the current study might be due to differences in the amount of information processing on the part of the infant and/or the “interestingness” of the toys.

Fifth, it is worth mentioning that, while infants gather information about their world through aggressive visual foraging, looking and attending are not synonymous. Previous research has shown that covert shifts of attention can occur without overt shifts of gaze by 4–6 months of age (72,73). However, the current study has focused exclusively on overt attention.

Finally, our laboratory setting was a novel environment for our participants and might have elicited behaviours that are different from the ones that develop at home. However, it still represents a significant advancement relative to other screen-based or highly controlled experimental tasks.

4. Materials and Methods

4.1 Experimental Design

Looking behaviour, EEG and ECG data were collected from mothers and their infants at two age points: 5 and 10 months while playing alone. At 5 months, infants were seated either in a highchair or on a researcher’s lap and a table was positioned in front so that toys on the table were within easy reach (see **Figure 1**). To reduce infant’s stress, mothers were present in the room but moved to another smaller table on the right side of the original table and given an identical set of toys which they played with in parallel. A wooden divider was positioned between the two tables to prevent infants from seeing the objects with which their mothers were playing. At 10 months, the same procedure was used but the divider was positioned across the midline of the table and the adult participants were seated directly opposite the infants. In both situations, mothers and infants had direct line of sight to one another but neither could see the others’ toys on the table.

The same three age-appropriate toys were always used for each age group. These were small and relatively engaging (see Figure S5). During the solo play interaction, one of the researchers sat behind the infant to collect the toys that fell on the floor (either because the infants threw them or because they fell from their hands) and brought them back on the table. Mothers were allowed to speak during the interaction but were instructed not to name the toys they were playing with to prevent infants from the influence of any exogenous parental’ influence. In the Supplementary Materials, we present a set of analyses that preclude the possibility of maternal influence on infants’ behaviour and demonstrate that the impact of the mothers on the infants’ behaviour did not differ between age groups (SI 3). The average duration of the interactions with usable EEG/ECG data did not differ significantly between 5 and 10 months (interactions with EEG (average duration at 5m = 292.4s, and 10m = 250.1s, $t(46) = -1.85$, $p = 0.07$); interactions with ECG data (average duration at 5m = 351.2s, and 10m = 317.9s, $t(40) = -1.1$, $p = 0.27$)).

The interactions were filmed using three Canon LEGRIA HF R806 camcorders recording at 50 fps. At 5 months, one camera was placed in front of the infant and another one was placed in front of the mother. At 10 months, two cameras faced the infant: one placed on the left of the divider, and one on the right. The other camera faced the mother and was positioned just behind the right side of the divider. All cameras were placed so that the infant’s and the mother’s gaze, as well as the three toys placed on the table, were always visible.

Brain activity was recorded using a 64-channel at 5 months and a 32-channel at 10 months, BioSemi gelbased ActiveTwo system with a sampling rate of 512Hz with no online filtering using Actiview Software.

Heart rate activity was recorded using a BioPac™ (Santa Barbara, CA) system recording at 2000Hz. ECG was recorded using disposable Ag-Cl electrodes placed in a lead II position.

4.2 Participants

Participants were typically developing infants and their mothers. The catchment area for this study was East London, including boroughs such as Tower Hamlets, Hackney and Newham. Participants were recruited postnatally through advertisements at local baby groups and local preschools/ nurseries. We also operated a word-of-mouth approach, asking parents who got involved to ask if their local networks would be interested in participating. Ethical approval was obtained from the University of East London ethics committee (application ID: ETH2021-0076). Informed consent, and consent to publish, was obtained by the caregivers of the infants tested.

Initial exclusion criteria included complex medical conditions (e.g., heart rate condition, neurological/ genetic abnormality), known developmental delays, prematurity, uncorrected vision difficulties, and parents below 18 years of age. Further exclusion criteria as well as final numbers of data included in each of the analyses for both samples are summarised in Table S7. The final sample included 12 infant females and 19 infant males at 5 months and 14 infant females and 15 infant males at 10 months. Data was analysed in a cross-sectional manner. Average age was 5.32 months (std = 0.58) and 10.49 months (std = 0.87). This is the first time that any of this data has been analysed and reported.

Since the analyses are performed conducted relative to specific events (such as the frequency and duration of looks to objects), and each participant averaged around 10 looks per minute at 5 months and approximately 4 looks at 10 months (see **Figure 1A and B**), we believe that the relatively low N for this study is balanced by the considerable amount of data points accessible.

4.3 Data processing and Statistical Analysis

4.3.1 Synchronisation between behavioural and EEG/ ECG data

The cameras were synchronised to the EEG and ECG via radio frequency (RF) receiver LED boxes attached to each camera. The RF boxes received trigger signals from a single source (computer running Matlab) at the beginning and end of the play session, and concurrently emitted light impulses, visible from each camera. At the same time, triggers were sent and stored in the Actiview Software and recorded to the EEG data as well as to the Acknowledge Software and recorded to the ECG data.

The video coding and EEG/ ECG data synchronisation was done by aligning the times of the LED lights and the EEG/ ECG triggers. We also checked for dropped/missing frames by checking that the time between the LED lights matched the times between the EEG/ ECG triggers.

4.3.2 Video coding

The looking behaviour of the infants was manually coded offline on a frame-by-frame basis, at 50fps. The start of a look was considered to be the first frame in which the gaze was static after moving to a new location. The following categories of gaze were coded: looks to objects (where the infant was focussing on one of the three objects), looks to partner (where the infant was looking at their partner), inattentive (where the infant was not looking to any of the objects nor the partner) and uncodable. Uncodable moments included periods where: 1) the infant's gaze was blocked or obscured by an object and/or their own hands, 2) their eyes were outside the camera frame, and/ or 3) a researcher was within the camera frame and the infant turned to them and/or realised a researcher was around. Video coding was completed by three coders, who were trained by the first author. To assess inter-rater reliability, ~15% of our data (10 datasets) were double-coded by a second coder and Cohen's kappa was calculated. There was moderate agreement (K = 0.581, std= 0.183) (74). Due to the unusual nature of our behavioural coding (with gaze coded across many 20ms bins) the interrater reliability is heavily contingent on how we calculate it. We chose to report the most stringent calculator of inter-rater reliability.

Looking behaviour data was then processed such that any look preceding and following an "uncodable" period was NaN-ed and excluded from further analyses. Similarly, both the first and the last look of every interaction were also NaN-ed and excluded from further analyses.

4.3.3 EEG artefact rejection and pre-processing

EEG data was pre-processed and cleaned from oculomotor and other contaminatory artefacts using a fully automatic artefact rejection procedure specially designed for naturalistic infant EEG data by Mariott Haresign (66), building on previous related work (59, 60). Briefly, this involved the following steps: 1) data were high-pass filtered at 1Hz, 2) line noise at 50Hz was eliminated using the EEGLAB function clean_line.m, 3) data were low-pass filtered at 20Hz, 4) the data were referenced to a robust average reference 5) noisy channels were rejected using the EEGLAB function pop_rejchan.m, 6) the channels identified in the previous stage were then interpolated back, using the EEGLAB function eeg_interp.m, 7) continuous data were automatically rejected (NaN-ed) in a sliding 1s epoch based on a percentage of bad channels (set here at 70% of channels) that exceed 5 standard deviations of the mean channel EEG power, and 8) Independent Component Analyses (ICA) were computed on the continuous data using the EEGLAB function runica.m. Only participants with fewer than 30% of channels interpolated at 5 months and 25% at 10 months (step 6) made it to the final step (step 8, ICA) and final analyses. To compare the quality of the EEG data at 5 and 10 months we performed a series of analyses on percentage of channels interpolated, total segments removed (i.e., zeroed out) and total ICA components rejected (see Table S5–6 and Figure S4).

The higher density array was down sampled so that all the EEG analyses described below used only shared channels between the 32- and the 64- channel EEG systems. We selected three main clusters of electrodes for our analyses: Frontal channels ('Fp1', 'Fp2', 'AF3', 'AF4', 'Fz'), Central channels ('FC1', 'FC2', 'C3', 'Cz', 'C4', 'CP1', 'CP2'), and Occipital channels ('PO3', PO4', O1', 'Oz', 'O1') (see Figure S6).

4.3.4 Heart Rate - Beats per minute

R-peak identification was performed using the in-built Matlab function 'findpeaks'. The minimum peak height was manually defined as a simple amplitude threshold after visualising the raw data, minimum peak distance, instead, was set at 230msec. Following this, automatic artefact rejection was performed by excluding those beats showing an inter-beat interval (IBI) <330 or >750msec (i.e., allowing a minimum of ~80BPM and maximum of ~180BPM), and by excluding those samples showing a rate of change of IBI greater than 90msec between samples. Next, we converted IBI values into beats-per-minute (BPM) values and removed outliers in the BPM time series. These were defined as values falling 2.5 interquartile ranges above the upper quartile and below the lower quartile. Outliers were then interpolated using the Matlab inbuilt function 'fillmissing' with the 'spline' method. Finally, we epoched the data into one-second epochs by averaging all the BPM values comprised in each one-second epoch.

4.3.5 Analysis 1. Developmental changes in attention

4.3.5.1 Overt attention and inattention extraction

The aim of our analysis was to identify moments where the infants paid attention to (i.e. looked at) any of the play objects as opposed to inattentive moments. Accordingly, all looks to object and inattentive looks were selected and categorised as attentional and inattentive episodes, respectively. Looks to partner were excluded from all analyses.

Following this, we extracted the first and last frame of all looks of interest (i.e., looks to objects and inattentive looks). To calculate attention and inattention durations, we subtracted the last frame from the first frame of each look of interest and divided it by the sampling rate (i.e., 50) to convert "duration in frames" to "duration in seconds". Attentive and inattentive reorientations (count) were calculated by counting the occurrence of each of these two attentional states and dividing it by the length of the session.

4.3.5.2 ACF of the attention duration

Here, we extracted the duration (in seconds) of all the attentional episodes that happened within the play session and zero-ed out all the non-attentional episodes. This allowed us to create a time series string with the durations of each consecutive attention episode. We then calculate the autocorrelation of that signal and repeated these steps for each behavioural dataset. Finally, we averaged the ACF values within each age group to obtain the ACF values reported in **Figure 3A**.

4.3.6 Analysis 2. Auto- and cross-correlation analyses between infant autonomic arousal and attention

4.3.6.1 Attention, one-second epochs

To calculate the "attention" variable, we epoched the gaze data into one-second epochs and calculated the duration (in msec) of each attention episode relative to the one-second epochs. Most epochs were coded as either 1000 (epochs where the child was attending throughout) or 0 (epochs where the child was inattentive throughout). If an attention episode started halfway through a one-second epoch, then it was coded as 500. The other non-attention episodes (i.e., inattentive looks, looks to partner) were zeroed. See **Figure 1** for a schematic view of the procedure we followed to parse the looking behaviour into one-second epochs.

4.3.6.2 Cross-correlation analyses

To investigate the relationship between autonomic arousal and fluctuations in attention we performed a cross-correlation analysis between the two variables. Importantly, these analyses are not time locked to specific moments (i.e., start of an attentional event) and are conducted on two time series (i.e., attention and heart rate fluctuations) as a whole. Because of this, the strength of the overall correlation is weakened by the fact that periods of expected stronger correlation are balanced by weaker correlations where we would not necessarily expect any correlation at all.

Additionally, we also computed the autocorrelation for autonomic arousal to assess how well it predicts itself over time and evaluate its stability. All analyses were computed at lags from -30 to +30s in 1s intervals. The cross-correlations values at each time-lag were computed individually and then averaged across all participants. The procedure was identical for the autocorrelation, except that instead of examining the relationship of two different time series at variable time intervals, we assessed the relationship of one time series to itself at variable time intervals.

To assess significance of the cross-correlations, we first used bootstrapping to generate confidence intervals, using an approach that controls for the level of autocorrelation in the data. To do this, the time series data of one participant (e.g., attention of participant 1) was randomly paired with the time series data of another participant (e.g., autonomic arousal of participant 13). If the time series datasets had different lengths (due to different participants having different session lengths), we appended zeros to the end of the shorter vector to match the length of the longest vector. We then computed the cross-correlation between all the unique combinations that could be found within each sample (e.g., in a sample N=23, the maximum of unique combinations is 529). Next, the cross-correlation results in the permuted data were randomly grouped in samples that were the same size as the original data (e.g., N= 23) and averaged together. This procedure was repeated 1000 times and used to generate the 95% confidence intervals. In this way, we identified whether the observed cross-correlation values at each time interval differed significantly from chance.

Next, to control for multiple comparisons across time intervals, we used a cluster-based permutation approach. On each iteration, one permutation was compared with the 999 other permutations, and significant time-points were identified as values falling above the 97.5th centile and below the 2.5th centile (corresponding to a significance level of 0.05). We then identified the two largest clusters of significance that occurred by chance: one for positive correlation values and the other for negative correlation values. We repeated this 1000 times. Following this, we created a distribution of cluster sizes for positive and negative correlation values and took the size value corresponding to the 95th percentile in each distribution to define our cluster-size threshold. Finally, we compared the cluster sizes obtained in the observed data against the cluster-size threshold and only considered significant the ones that exceeded such threshold.

Calculating the significance levels of the autocorrelation was more straightforward. This was done by first calculating the autocorrelations based on individual datasets, and then averaging the significance values of the Spearman's correlations at each time interval.

4.3.7 Analysis 3. Calculation of neural power changes around an attention episode

We examined the associations between the duration of infant attention episodes and infant theta changes around these looks using linear-mixed effects models. Infant attention episodes and attention duration were calculated as explained above in section 4.3.5.1.

To conduct these analyses, each infant attention episode onset (i.e., gaze shift to a different toy) was identified in the EEG signal by calculating the time from the start of the interaction (first LED) to the onset of the look (in the behavioural data) and adding it to the first EEG trigger. For each look, we extracted theta (3–6Hz) power for two time-windows immediately prior to the onset of each look (-2000 to -1000msec and -1000 to 0 msec pre-look onset) and three time-windows immediately after the onset of each look (0 to 1000ms, 1000ms to 2000msecand 2000 to 3000msec post look onset) (see **Figure 1**). To calculate the EEG power spectra, we use the 'mtmfft' method from the ft_freqanalysis function in FieldTrip, an open-source Matlab toolbox (75); http://fieldtriptoolbox.org). Extreme power values that were 4 times greater than the interquartile range were treated as outliers and excluded from further analyses (similar to 29). More detail on the amount of data available (i.e., average duration of the session per participant and number and duration of attentional episodes per minute) can be accessed in Table S1.

For each epoch, we only selected power within our cluster of central channels (similar to 60). Power at each bin was expressed as relative power, defined as the total power at a specific frequency band (e.g. 3 to 6Hz for theta) divided by the total power across all frequency bands (1 to 20Hz) during that epoch. After extracting the relative power in the theta band, we calculated separate linear mixed effects models for each of the five windows to examine the relationship between EEG power within that time window and attention duration.

4.3.8 Analysis 4. Calculation of neural power changes within an attention episode

In addition, we also wanted to look at power changes within attention episodes. Infant attention episodes and attention duration were calculated as explained above (section 4.3.5.1) and each infant look onset towards an object was identified in the EEG signal as described in analysis 3 (section 4.3.7). For each look, we extracted the first (0 to +1000msec, "start") and third-to-fourth (3000 to 4000msec, "middle") second into the look, and the last second (-1000msec prior to look offset to look offset, “termination”) before look termination (see **Figure 1**). Looks that did not make it to the full second segment were excluded from further analysis. Similarly, only looks that were longer than 5 seconds were included to the “middle” group. This was done to avoid an overlap between the activity from the “middle” and the “termination” groups.

Frequency analysis was conducted to assess the power spectral density for both theta (3–6Hz) and alpha (69Hz) frequency rhythms for each of the three time-segments. These analyses were calculated for the three prespecified clusters of channels: Frontal, Central and Occipital (see Figure S6). Again, power at each time segment was expressed as relative power.

The selection of both theta (3–6Hz) and alpha (6–9Hz) frequency bands was led by previous work using this same approach (e.g., 28, 29, 30, 37).

4.3.9 Analysis 5. Auto- and cross-correlations analyses between infant theta activity and attention

4.3.9.1 EEG relative power, one-second epochs

For this analysis, we parsed the EEG data into one-second segments and calculated the relative theta power for each one-second segment as described above (see 4.3.7).

4.3.9.2 Cross-correlation analysis

To explore whether modulations in endogenous theta activity related to fluctuations in infants’ attention, we conducted a cross-correlation analysis between infants’ relative theta and attention. Attention was calculated as described in analysis 2 (4.3.6.1, and **Figure 1**). Additionally, we also computed the autocorrelation for relative theta to assess how theta predicts itself over time. Again, all analyses were computed at lags from -30 to +30s in 1s intervals. Significance was assessed following the steps described in analysis 2 (4.3.6.2).

Finally, to explore interdependencies between autonomic arousal and theta activity we conducted a crosscorrelation analysis between infants’ autonomic activity and relative theta.

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Ethics

Human subjects: Ethical approval was obtained from the University of East London ethics committee (application ID: ETH2021-0076). Informed consent, and consent to publish, was obtained by the caregivers of the infants tested. Specific informed consent was also obtained from the caregivers of the infants featured in **Figure 1**.

Author contributions

Conceptualization: SVW, EJHJ, MPA.

Data curation: IMH, MW, MPA.

Formal analysis: MPA.

Funding acquisition: SVW.

Investigation: EG, JI, PL, IMH, EAMP, NKV, MW, MPA.

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Project administration: EG, JI, PL, IMH, TJN, EAMP, NKV, MW, SVW, MPA.

Supervision: SVW, EJHJ.

Visualisation: MPA.

Writing – original draft: MPA.

Writing – review & editing: EG, JI, PL, IMH, TJN, EAMP, NKV, MW, EJHJ, SVW, MPA.

Competing interests

Authors declare that they have no competing interests.

Data and materials availability

Partial restrictions to the data and/or materials apply. Due to the personally identifiable nature of this data (video recordings from infants) the raw data will not be made publicly accessible. Researchers who wish to access the raw data should email the lead author. Permission to access the raw data will be granted as long as the applicant can guarantee that certain privacy guidelines (e.g. storing the data only on secure, encrypted servers, and a guarantee not to share it with anyone else) can be provided.

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Editors

Reviewing Editor

Marius Peelen

Radboud University Nijmegen, Nijmegen, Netherlands

Senior Editor

Yanchao Bi

Beijing Normal University, Beijing, China

Reviewer #1 (Public Review):

Summary:

The paper investigates the physiological and neural processes that relate to infants' attention allocation in a naturalistic setting. Contrary to experimental paradigms that are usually employed in developmental research, this study investigates attention processes while letting the infants free to play with three toys in the vicinity of their caregiver, which is closer to a common, everyday life context. The paper focuses on infants at 5 and 10 months of age and finds differences in what predicts attention allocation. At 5 months, attention episodes are shorter and their duration is predicted by autonomic arousal. At 10 months, attention episodes are longer, and their duration can be predicted by theta power. Moreover, theta power predicted the proportion of looking at the toys, as well as a decrease in arousal (heart rate). Overall, the authors conclude that attentional systems change across development, becoming more driven by cortical processes.

Strengths:

I enjoyed reading the paper, I am impressed with the level of detail of the analyses, and I am strongly in favour of the overall approach, which tries to move beyond in-lab settings. The collection of multiple sources of data (EEG, heart rate, looking behaviour) at two different ages (5 and 10 months) is a key strength of this paper. The original analyses, which build onto robust EEG preprocessing, are an additional feat that improves the overall value of the paper. The careful consideration of how theta power might change before, during, and in the prediction of attention episodes is especially remarkable.

Weaknesses:

The levels of EEG noise across age groups and periods of attention allocation are not controlled for. I appreciate the analysis of noise reported in supplementary materials. The analysis focuses on a broad level (average noise in 5-month-olds vs 10-month-olds) but variations might be more fine-grained (for example, noise in 5mos might be due to fussiness and crying, while at 10 months it might be due to increased movements). More importantly, noise might even be the same across age groups, but correlated to other aspects of their behaviour (head or eye movements) that are directly related to the measures of interest. Is it possible that noise might co-vary with some of the behaviours of interest, thus leading to either spurious effects or false negatives? One way to address this issue would be for example to check if noise in the signal can predict attention episodes. If this is the case, noise should be added as a covariate in many of the analyses of this paper.

Concerning cross-correlation analyses, the authors state that "Interpreting the exact time intervals over which a cross-correlation is significant is challenging". Then, they say that asymmetry is enough to conclude that attention forward predicted theta power more than vice versa. I think it could be useful to add a bit more of explanation before reaching this conclusion, explaining why such statement is correct, and how it is supported by previous work in statistics.

Finally, the cognitive process under investigation (e.g., attention) and its operationalization (e.g., duration of consecutive looking toward a toy) are not fully distinguished, but conflated instead (e.g., "attention durations"). This does not impact the quality of the work or analyses, but it slightly reduces clarity.

General Remarks
In general, the authors achieved their aim in that they successfully showed the relationship between looking behaviour (as a proxy of attention), autonomic arousal, and electrophysiology. Two aspects are especially interesting. First, the fact that at 5 months, autonomic arousal predicts the duration of subsequent attention episodes, but at 10 months this effect is not present. Conversely, at 10 months, theta power predicts the duration of looking episodes, but this effect is not present in 5-month-old infants. This pattern of results suggests that younger infants have less control over their attention, which mostly depends on their current state of arousal, but older infants have gained cortical control of their attention, which in turn impacts their looking behaviour and arousal.

https://doi.org/10.7554/eLife.92171.2.sa1

Reviewer #2 (Public Review):

Summary:

This manuscript explores infants' attention patterns in real-world settings and their relationship with autonomic arousal and EEG oscillations in the theta frequency band. The study included 5- and 10-month-old infants during free play. The results showed that the 5-month-old group exhibited a decline in HR forward-predicted attentional behaviors, while the 10-month-old group exhibited increased theta power following shifts in gaze, indicating the start of a new attention episode. Additionally, this increase in theta power predicted the duration of infants' looking behavior.

Strengths:

The study's strengths lie in its utilization of advanced protocols and cutting-edge techniques to assess infants' neural activity and autonomic arousal associated with their attention patterns, as well as the extensive data coding and processing. Overall, I think this article's findings have important theoretical implications for the development of infant attention.

Weaknesses:

The authors have effectively tackled the majority of my concerns within their revised manuscript, resulting in a substantial improvement. While the revised paper notably addresses many points, one question regarding the potential contamination of saccades on EEG power remains partially unresolved. However, I appreciate the authors' explanation that resolving this issue was challenging due to the absence of eye-tracking data in the current study. Additionally, I acknowledge their inclusion of this concern in the limitations section.

https://doi.org/10.7554/eLife.92171.2.sa0

Author response:

The following is the authors' response to the previous reviews.

> ***eLife assessment***
>
> *This important study explores infants' attention patterns in real-world settings using advanced protocols and cutting-edge methods. The presented evidence for the role of EEG theta power in infants' attention is currently incomplete. The study will be of interest to researchers working on the development and control of attention.*
>
> ***Public Reviews:***
>
> ***Reviewer #1 (Public Review):***
>
> *Summary:*
>
> *The paper investigates the physiological and neural processes that relate to infants' attention allocation in a naturalistic setting. Contrary to experimental paradigms that are usually employed in developmental research, this study investigates attention processes while letting the infants be free to play with three toys in the vicinity of their caregiver, which is closer to a common, everyday life context. The paper focuses on infants at 5 and 10 months of age and finds differences in what predicts attention allocation. At 5 months, attention episodes are shorter and their duration is predicted by autonomic arousal. At 10 months, attention episodes are longer, and their duration can be predicted by theta power. Moreover, theta power predicted the proportion of looking at the toys, as well as a decrease in arousal (heart rate). Overall, the authors conclude that attentional systems change across development, becoming more driven by cortical processes.*

Strengths:

I enjoyed reading the paper, I am impressed with the level of detail of the analyses, and I am strongly in favour of the overall approach, which tries to move beyond in-lab settings. The collection of multiple sources of data (EEG, heart rate, looking behaviour) at two different ages (5 and 10 months) is a key strength of this paper. The original analyses, which build onto robust EEG preprocessing, are an additional feat that improves the overall value of the paper. The careful consideration of how theta power might change before, during, and in the prediction of attention episodes is especially remarkable. However, I have a few major concerns that I would like the authors to address, especially on the methodological side.

Points of improvement

(1) Noise

The first concern is the level of noise across age groups, periods of attention allocation, and metrics. Starting with EEG, I appreciate the analysis of noise reported in supplementary materials. The analysis focuses on a broad level (average noise in 5-month-olds vs 10-month-olds) but variations might be more fine-grained (for example, noise in 5mos might be due to fussiness and crying, while at 10 months it might be due to increased movements). More importantly, noise might even be the same across age groups, but correlated to other aspects of their behaviour (head or eye movements) that are directly related to the measures of interest. Is it possible that noise might co-vary with some of the behaviours of interest, thus leading to either spurious effects or false negatives? One way to address this issue would be for example to check if noise in the signal can predict attention episodes. If this is the case, noise should be added as a covariate in many of the analyses of this paper.

We thank the reviewer for this comment. We certainly have evidence that even the most state-of-the-art cleaning procedures (such as machine-learning trained ICA decompositions, as we applied here) are unable to remove eye movement artifact entirely from EEG data (Haresign et al., 2021; Phillips et al., 2023). (This applies to our data but also to others' where confounding effects of eye movements are generally not considered.) Importantly, however, our analyses have been designed very carefully with this explicit challenge in mind. All of our analyses compare changes in the relationship between brain activity and attention as a function of age, and there is no evidence to suggest that different sources of noise (e.g. crying vs. movement) would associate differently with attention durations nor change their interactions with attention over developmental time. And figures 5 and 7, for example, both look at the relationship of EEG data at one moment in time to a child's attention patterns hundreds or thousands of milliseconds before and after that moment, for which there is no possibility that head or eye movement artifact can have systematically influenced the results.

Moving onto the video coding, I see that inter-rater reliability was not very high. Is this due to the fine-grained nature of the coding (20ms)? Is it driven by differences in expertise among the two coders? Or because coding this fine-grained behaviour from video data is simply too difficult? The main dependent variable (looking duration) is extracted from the video coding, and I think the authors should be confident they are maximising measurement accuracy.

We appreciate the concern. To calculate IRR we used this function (Cardillo G. (2007) Cohen's kappa: compute the Cohen's kappa ratio on a square matrix. http://www.mathworks.com/matlabcentral/fileexchange/15365). Our "Observed agreement" was 0.7 (std= 0.15). However, we decided to report the Cohen's kappa coefficient, which is generally thought to be a more robust measure as it takes into account the agreement occurring by chance. We conducted the training meticulously (refer to response to Q6, R3), and we have confidence that our coders performed to the best of their abilities.

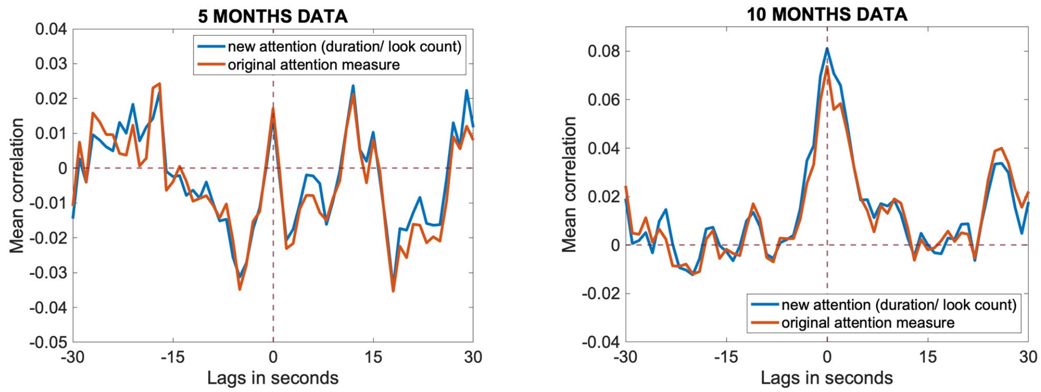
> *(2) Cross-correlation analyses*
>
> *I would like to raise two issues here. The first is the potential problem of using auto-correlated variables as input for cross-correlations. I am not sure whether theta power was significantly autocorrelated. If it is, could it explain the cross-correlation result? The fact that the cross-correlation plots in Figure 6 peak at zero, and are significant (but lower) around zero, makes me think that it could be a consequence of periods around zero being autocorrelated. Relatedly: how does the fact that the significant lag includes zero, and a bit before, affect the interpretation of this effect?*

Just to clarify this analysis, we did include a plot showing autocorrelation of theta activity in the original submission (Figs 7A and 7B in the revised paper). These indicate that theta shows little to no autocorrelation. And we can see no way in which this might have influenced our results. From their comments, the reviewer seems rather to be thinking of phasic changes in the autocorrelation, and whether the possibility that greater stability in theta during the time period around looks might have caused the cross-correlation result shown in 7E. Again though we can see no way in which this might be true, as the cross-correlation indicates that greater theta power is associated with a greater likelihood of looking, and this would not have been affected by changes in the autocorrelation.

> *A second issue with the cross-correlation analyses is the coding of the looking behaviour. If I understand correctly, if an infant looked for a full second at the same object, they would get a maximum score (e.g., 1) while if they looked at 500ms at the object and 500ms away from the object, they would receive a score of e.g., 0.5. However, if they looked at one object for 500ms and another object for 500ms, they would receive a maximum score (e.g., 1). The reason seems unclear to me because these are different attention episodes, but they would be treated as one. In addition, the authors also show that within an attentional episode theta power changes (for 10mos). What is the reason behind this scoring system? Wouldn't it be better to adjust by the number of attention switches, e.g., with the formula: looking-time/(1+N_switches), so that if infants looked for a full second, but made 1 switch from one object to the other, the score would be .5, thus reflecting that attention was terminated within that episode?*

We appreciate this suggestion. This is something we did not consider, and we thank the reviewer for raising it. In response to their comment, we have now rerun the analyses using the new measure (looking-time/(1+N_switches), and we are reassured to find that the results remain highly consistent. Please see Author response image 1 below where you can see the original results in orange and the new measure in blue at 5 and 10 months.

Author response image 1.



> *(3) Clearer definitions of variables, constructs, and visualisations*
>
> *The second issue is the overall clarity and systematicity of the paper. The concept of attention appears with many different names. Only in the abstract, it is described as attention control, attentional behaviours, attentiveness, attention durations, attention shifts and attention episode. More names are used elsewhere in the paper. Although some of them are indeed meant to describe different aspects, others are overlapping. As a consequence, the main results also become more difficult to grasp. For example, it is stated that autonomic arousal predicts attention, but it's harder to understand what specific aspect (duration of looking, disengagement, etc.) it is predictive of. Relatedly, the cognitive process under investigation (e.g., attention) and its operationalization (e.g., duration of consecutive looking toward a toy) are used interchangeably. I would want to see more demarcation between different concepts and between concepts and measurements.*

We appreciate the comment and we have clarified the concepts and their operationalisation throughout the revised manuscript.

> *General Remarks*
>
> *In general, the authors achieved their aim in that they successfully showed the relationship between looking behaviour (as a proxy of attention), autonomic arousal, and electrophysiology. Two aspects are especially interesting. First, the fact that at 5 months, autonomic arousal predicts the duration of subsequent attention episodes, but at 10 months this effect is not present. Conversely, at 10 months, theta power predicts the duration of looking episodes, but this effect is not present in 5-month-old infants. This pattern of results suggests that younger infants have less control over their attention, which mostly depends on their current state of arousal, but older infants have gained cortical control of their attention, which in turn impacts their looking behaviour and arousal.*

We thank the reviewer for the close attention that they have paid to our manuscript, and for their insightful comments.

> ***Reviewer #2 (Public Review):***
>
> *Summary:*

This manuscript explores infants' attention patterns in real-world settings and their relationship with autonomic arousal and EEG oscillations in the theta frequency band. The study included 5- and 10-month-old infants during free play. The results showed that the 5-month-old group exhibited a decline in HR forward-predicted attentional behaviors, while the 10-month-old group exhibited increased theta power following shifts in gaze, indicating the start of a new attention episode. Additionally, this increase in theta power predicted the duration of infants' looking behavior.

Strengths:

The study's strengths lie in its utilization of advanced protocols and cutting-edge techniques to assess infants' neural activity and autonomic arousal associated with their attention patterns, as well as the extensive data coding and processing. Overall, the findings have important theoretical implications for the development of infant attention.

Weaknesses:

Certain methodological procedures require further clarification, e.g., details on EEG data processing. Additionally, it would be beneficial to eliminate possible confounding factors and consider alternative interpretations, e,g., whether the differences observed between the two age groups were partly due to varying levels of general arousal and engagement during the free play.

We thank the reviewer for their suggestions and have addressed them in our point-by-point responses below.

Reviewer #3 (Public Review):

Summary:

Much of the literature on attention has focused on static, non-contingent stimuli that can be easily controlled and replicated--a mismatch with the actual day-to-day deployment of attention. The same limitation is evident in the developmental literature, which is further hampered by infants' limited behavioral repertoires and the general difficulty in collecting robust and reliable data in the first year of life. The current study engages young infants as they play with age-appropriate toys, capturing visual attention, cardiac measures of arousal, and EEG-based metrics of cognitive processing. The authors find that the temporal relations between measures are different at age 5 months vs. age 10 months. In particular, at 5 months of age, cardiac arousal appears to precede attention, while at 10 months of age attention processes lead to shifts in neural markers of engagement, as captured in theta activity.

Strengths:

The study brings to the forefront sophisticated analytical and methodological techniques to bring greater validity to the work typically done in the research lab. By using measures in the moment, they can more closely link biological measures to actual behaviors and cognitive stages. Often, we are forced to capture these measures in separate contexts and then infer in-the-moment relations. The data and techniques provide insights for future research work.

Weaknesses:

The sample is relatively modest, although this is somewhat balanced by the sheer number of data points generated by the moment-to-moment analyses. In addition, the study is cross-sectional, so the data cannot capture true change over time. Larger samples, followed over time, will provide a stronger test for the robustness and reliability of the preliminary data noted here. Finally, while the method certainly provides for a more active and interactive infant in testing, we are a few steps removed from the complexity of daily life and social interactions.

We thank the reviewer for their suggestions and have addressed them in our point-by-point responses below.

Reviewer #1 (Recommendations For The Authors):

Here are some specific ways in which clarity can be improved:

A. Regarding the distinction between constructs, or measures and constructs:

i. In the results section, I would prefer to mention looking at duration and heart rate as metrics that have been measured, while in the introduction and discussion, a clear 1-to-1 link between construct/cognitive process and behavioural or (neuro)psychophysical measure can be made (e.g., sustained attention is measured via looking durations; autonomic arousal is measured via heart-rate).

The way attention and arousal were operationalised are now clarified throughout the text, especially in the results.

ii. Relatedly, the "attention" variable is not really measuring attention directly. It is rather measuring looking time (proportion of looking time to the toys?), which is the operationalisation, which is hypothesised to be related to attention (the construct/cognitive process). I would make the distinction between the two stronger.

This distinction between looking and paying attention is clearer now in the reviewed manuscript as per R1 and R3's suggestions. We have also added a paragraph in the Introduction to clarify it and pointed out its limitations (see pg.5).

B. Each analysis should be set out to address a specific hypothesis. I would rather see hypotheses in the introduction (without direct reference to the details of the models that were used), and how a specific relation between variables should follow from such hypotheses. This would also solve the issue that some analyses did not seem directly necessary to the main goal of the paper. For example:

i. Are ACF and survival probability analyses aimed at proving different points, or are they different analyses to prove the same point? Consider either making clearer how they differ or moving one to supplementary materials.

We clarified this in pg. 4 of the revised manuscript.

ii. The autocorrelation results are not mentioned in the introduction. Are they aiming to show that the variables can be used for cross-correlation? Please clarify their role or remove them.

We clarified this in pg. 4 of the revised manuscript.

C. Clarity of cross-correlation figures. To ensure clarity when presenting a cross-correlation plot, it's important to provide information on the lead-lag relationships and which variable is considered X and which is Y. This could be done by labelling the axes more clearly (e.g., the left-hand side of the - axis specifies x leads y, right hand specifies y leads x) or adding a legend (e.g., dashed line indicates x leading y, solid line indicates y

> *leading x). Finally, the limits of the x-axis are consistent across plots, but the limits of the y-axis differ, which makes it harder to visually compare the different plots. More broadly, the plots could have clearer labels, and their resolution could also be improved.*

This information on what variable precedes/ follows was in the caption of the figures. However, we have edited the figures as per the reviewer's suggestion and added this information in the figures themselves. We have also uploaded all the figures in higher resolution.

> *D. Figure 7 was extremely helpful for understanding the paper, and I would rather have it as Figure 1 in the introduction.*

We have moved figure 7 to figure 1 as per this request.

> *E. Statistics should always be reported, and effects should always be described. For example, results of autocorrelation are not reported, and from the plot, it is also not clear if the effects are significant (the caption states that red dots indicate significance, but there are no red dots. Does this mean there is no autocorrelation?).*

We apologise – this was hard to read in the original. We have clarified that there is no autocorrelation present in Fig 7A and 7D.

> *And if so, given that theta is a wave, how is it possible that there is no autocorrelation (connected to point 1)?*

We thank the reviewer for raising this point. In fact, theta power is looking at oscillatory activity in the EEG within the 3-6Hz window (i.e. 3 to 6 oscillations per second). Whereas we were analysing the autocorrelation in the EEG data by looking at changes in theta power between consecutive 1 second long windows. To say that there is no autocorrelation in the data means that, if there is more 3-6Hz activity within one particular 1-second window, there tends not to be significantly more 3-6Hz activity within the 1-second windows immediately before and after.

> *F. Alpha power is introduced later on, and in the discussion, it is mentioned that the effects that were found go against the authors' expectations. However, alpha power and the authors' expectations about it are not mentioned in the introduction.*

We thank the reviewer for this comment. We have added a paragraph on alpha in the introduction (pg.4).

> *Minor points:*
>
> *1. At the end of 1st page of introduction, the authors state that:*
>
> *"How children allocate their attention in experimenter-controlled, screen-based lab tasks differs, however, from actual real-world attention in several ways (32-34). For example, the real-world is interactive and manipulable, and so how we interact with the world determines what information we, in turn, receive from it: experiences generate behaviours (35)."*
>
> *I think there's more to this though - Lab-based studies can be made interactive too (e.g., Meyer et al., 2023, Stahl & Feigenson, 2015). What remains unexplored is how infants actively and freely initiate and self-structure their attention, rather than how they respond to experimental manipulations.*

> *Meyer, M., van Schaik, J. E., Poli, F., & Hunnius, S. (2023). How infant-directed actions enhance infants' attention, learning, and exploration: Evidence from EEG and computational modeling. Developmental Science, 26(1), e13259.*
>
> *Stahl, A. E., & Feigenson, L. (2015). Observing the unexpected enhances infants' learning and exploration. Science, 348(6230), 91-94.*

We thank the reviewer for this suggestion and added their point in pg. 4.

> *(2) Regarding analysis 4:*
>
> *a. In analysis 1 you showed that the duration of attentional episodes changes with age. Is it fair to keep the same start, middle, and termination ranges across age groups? Is 3-4 seconds "middle" for 5-month-olds?*

We appreciate the comment. There are many ways we could have run these analyses and, in fact, in other papers we have done it differently, for example by splitting each look in 3, irrespective of its duration (Phillips et al., 2023).

However, one aspect we took into account was the observation that 5-month-old infants exhibited more shorter looks compared to older infants. We recognized that dividing each into 3 parts, regardless of its duration, might have impacted the results. Presumably, the activity during the middle and termination phases of a 1.5-second look differs from that of a look lasting over 7 seconds.

Two additional factors that provided us with confidence in our approach were: 1) while the definition of "middle" was somewhat arbitrary, it allowed us to maintain consistency in our analyses across different age points. And, 2) we obtained a comparable amount of observations across the two time points (e.g. “middle” at 5 months we had 172 events at 5 months, and 194 events at 10 months).

> *b. It is recommended not to interpret lower-level interactions if more complex interactions are not significant. How are the interaction effects in a simpler model in which the 3-way interaction is removed?*

We appreciate the comment. We tried to follow the same steps as in (Xie et al., 2018). However, we have re-analysed the data removing the 3-way interaction and the significance of the results stayed the same. Please see Author response image 2 below (first: new analyses without the 3-way interactions, second: original analyses that included the 3-way interaction).

Author response image 2.

Analysis of Variance

Source	Sum Sq.	d.f.	Mean Sq.	F	Prob>F
attentional phase	0.174	2	0.08719	4.56	0.0105
channel cluster	0.043	2	0.0216	1.13	0.3232
age	0.022	1	0.02245	1.17	0.2787
attentional phase:channel cluster	0.013	4	0.00318	0.17	0.9557
attentional phase:age	0.213	2	0.10668	5.58	0.0038
channel cluster:age	0.564	2	0.28184	14.74	0
Error	136.571	7141	0.01912		
Total	137.741	7154			

Analysis of Variance

Source	Sum Sq.	d.f.	Mean Sq.	F	Prob>F
attentional phase	0.174	2	0.08719	4.56	0.0105
channel cluster	0.045	2	0.02255	1.18	0.3077
age	0.022	1	0.02245	1.17	0.2788
attentional phase*channel cluster	0.019	4	0.00478	0.25	0.9098
attentional phase*age	0.213	2	0.10668	5.58	0.0038
channel cluster*age	0.458	2	0.22918	11.98	0
attentional phase*channel cluster*age	0.026	4	0.00644	0.34	0.8535
Error	136.545	7137	0.01913		
Total	137.741	7154			

Constrained (Type III) sums of squares.

> *(3) Figure S1: there seems to be an outlier in the bottom-right panel. Do results hold excluding it?*

We re-run these analyses as per this suggestion and the results stayed the same (refer to SM pg. 2).

> *(4) Figure S2 should refer to 10 months instead of 12.*

We thank the reviewer for noticing this typo, we have changed it in the reviewed manuscript (see SM pg. 3).

> *(5) In the 2nd paragraph of the discussion, I found this sentence unclear: "From Analysis 1 we found that infants at both ages showed a preferred modal reorientation rate".*

We clarified this in the reviewed manuscript in pg10

> *(6) Discussion: many (infant) studies have used theta in anticipation of receiving information (Begus et al., 2016) surprising events (Meyer et al., 2023), and especially exploration (Begus et al., 2015). Can you make a broader point on how these findings inform our interpretation of theta in the infant population (go more from description to underlying mechanisms)?*

We have extended on this point on interpreting frequency bands in pg13 of the reviewed manuscript and thank the reviewer for bringing it up.

Begus, K., Gliga, T., & Southgate, V. (2016). Infants' preferences for native speakers are associated with an expectation of information. Proceedings of the National Academy of Sciences, 113(44), 12397-12402.

Meyer, M., van Schaik, J. E., Poli, F., & Hunnius, S. (2023). How infant-directed actions enhance infants' attention, learning, and exploration: Evidence from EEG and computational modeling. Developmental Science, 26(1), e13259.

Begus, K., Southgate, V., & Gliga, T. (2015). Neural mechanisms of infant learning: differences in frontal theta activity during object exploration modulate subsequent object recognition. Biology letters, 11(5), 20150041.

> *(7) 2nd page of discussion, last paragraph: "preferred modal reorientation timer" is not a neural/cognitive mechanism, just a resulting behaviour.*
>
> *We agree with this comment and thank the reviewer for bringing it out to our attention. We clarified this in in pg12 and pg13 of the reviewed manuscript.*
>
> ***Reviewer #2 (Recommendations For The Authors):***
>
> *I have a few comments and questions that I think the authors should consider addressing in a revised version. Please see below:*
>
> *(1) During preprocessing (steps 5 and 6), it seems like the "noisy channels" were rejected using the pop_rejchan.m function and then interpolated. This procedure is common in infant EEG analysis, but a concern arises: was there no upper limit for channel interpolation? Did the authors still perform bad channel interpolation even when more than 30% or 40% of the channels were identified as "bad" at the beginning with the continuous data?*

We did state in the original manuscript that "participants with fewer than 30% channels interpolated at 5 months and 25% at 10 months made it to the final step (ICA) and final analyses". In the revised version we have re-written this section in order to make this more clear (pg. 17).

> *(2) I am also perplexed about the sequencing of the ICA pruning step. If the intention of ICA pruning is to eliminate artificial components, would it be more logical to perform this procedure before the conventional artifacts' rejection (i.e., step 7), rather than after? In addition, what was the methodology employed by the authors to identify the artificial ICA components? Was it done through manual visual inspection or utilizing specific toolboxes?*

We agree that the ICA is often run before, however, the decision to reject continuous data prior to ICA was to remove the very worst sections of data (where almost all channels were affected), which can arise during times when infants fuss or pull the caps. Thus, this step was applied at this point in the pipeline so that these sections of really bad data were not inputted into the ICA. This is fairly widespread practice in cleaning infant data.

Concerning the reviewer's second question, of how ICA components were removed – the answer to this is described in considerable detail in the paper that we refer to in that setion of the manuscript. This was done by training a classifier specially designed to clean naturalistic infant EEG data (Haresign et al., 2021) and has since been employed in similar studies (e.g. Georgieva et al., 2020; Phillips et al., 2023).

> *(3) Please clarify how the relative power was calculated for the theta (3-6Hz) and alpha (6-9Hz) bands. Were they calculated by dividing the ratio of theta or alpha power to the power between 3 and 9Hz, or the total power between 1 (or 3) and 20 Hz? In other words, what does the term "all frequency bands" refer to in section 4.3.7?*

We thank the reviewer for this comment, we have now clarified this in pg. 22.

> *(4) One of the key discoveries presented in this paper is the observation that attention shifts are accompanied by a subsequent enhancement in theta band power shortly after the shifts occur. Is it possible that this effect or alteration might be linked to infants' saccades, which are used as indicators of attention shifts? Would it be feasible to analyze the disparities in amplitude between the left and right frontal electrodes (e.g., Fp1 and Fp2, which could be viewed as virtual horizontal EOG channels) in relation to theta band power, in order to eliminate the possibility that the augmentation of theta power was attributable to the intensity of the saccades?*

We appreciate the concern. Average saccade duration in infants is about 40ms (Garbutt et al., 2007). Our finding that the positive cross-correlation between theta and look duration is present not only when we examine zero-lag data but also when we examine how theta forwards-predicts attention 1-2 seconds afterwards seems therefore unlikely to be directly attributable to saccade-related artifact. Concerning the reviewer's suggestion – this is something that we have tried in the past. Unfortunately, however, our experience is that identifying saccades based on the disparity between Fp1 and Fp2 is much too unreliable to be of any use in analysing data. Even if specially positioned HEOG electrodes are used, we still find the saccade detection to be insufficiently reliable. In ongoing work we are tracking eye movements separately, in order to be able to address this point more satisfactorily.

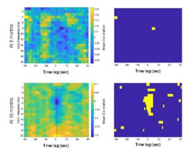
> *(5) The following question is related to my previous comment. Why is the duration of the relationship between theta power and moment-to-moment changes in attention so short? If theta is indeed associated with attention and information processing, shouldn't the relationship between the two variables strengthen as the attention episode progresses? Given that the authors themselves suggest that "One possible interpretation of this is that neural activity associates with the maintenance more than the initiation of attentional behaviors," it raises the question of (is in contradiction to) why the duration of the relationship is not longer but declines drastically (Figure 6).*

We thank the reviewer for raising this excellent point. Certainly we argue that this, together with the low autocorrelation values for theta documented in Fig 7A and 7D challenge many conventional ways of interpreting theta. We are continuing to investigate this question in ongoing work.

> *(6) Have the authors conducted a comparison of alpha relative power and HR deceleration durations between 5 and 10-month-old infants? This analysis could provide insights into whether the differences observed between the two age groups were partly due to varying levels of general arousal and engagement during free play.*

We thank the reviewer for this suggestion. Indeed, this is an aspect we investigated but ultimately, given that our primary emphasis was on the theta frequency, and considering the length of the manuscript, we decided not to incorporate. However, we attached Author response image 3 below showing there was no significant interaction between HR and alpha band.

Author response image 3.



> ***Reviewer #3 (Recommendations For The Authors):***
>
> *(1) In reading the manuscript, the language used seems to imply longitudinal data or at the very least the ability to detect change or maturation. Given the cross-sectional nature of the data, the language should be tempered throughout. The data are illustrative but not definitive.*

We thank the reviewer for this comment. We have now clarified that "Data was analysed in a cross-sectional manner" in pg15.

> *(2) The sample size is quite modest, particularly in the specific age groups. This is likely tempered by the sheer number of data points available. This latter argument is implied in the text, but not as explicitly noted. (However, I may have missed this as the text is quite dense). I think more notice is needed on the reliability and stability of the findings given the sample.*

We have clarified this in pg16.

> *(3) On a related note, how was the sample size determined? Was there a power analysis to help guide decision-making for both recruitment and choosing which analyses to proceed with? Again, the analytic approach is quite sophisticated and the questions are of central interest to researchers, but I was left feeling maybe these two aspects of the study were out-sprinting the available data. The general impression is that the sample is small, but it is not until looking at table s7, that it is in full relief. I think this should be more prominent in the main body of the study.*

We have clarified this in pg16.

> *(4) The devotes a few sentences to the relation between looking and attention. However, this distinction is central to the design of the study, and any philosophical differences regarding what take-away points can be generated. In my reading, I think this point needs to be more heavily interrogated.*

This distinction between looking and paying attention is clearer now in the reviewed manuscript as per R1 and R3's suggestions. We have also added a paragraph in the Introduction to clarify it and pointed out its limitations (see pg.5).

> *(5) I would temper the real-world attention language. This study is certainly a great step forward, relative to static faces on a computer screen. However, there are still a great number of artificial constraints that have been added. That is not to say that the constraints are bad--they are necessary to carry out the work. However, it should be acknowledged that it constrains the external validity.*

We have added a paragraph to acknowledged limitations of the setup in pg. 14.

> *(6) The kappa on the coding is not strong. The authors chose to proceed nonetheless. Given that, I think more information is needed on how coders were trained, how they were standardized, and what parameters were used to decide they were ready to code independently. Again, with the sample size and the kappa presented, I think more discussion is needed regarding the robustness of the findings.*

We appreciate the concern. As per our answer to R1, we chose to report the most stringent calculator of inter-rater reliability, but other calculation methods (i.e., percent agreement) return higher scores (see response to R1).

As per the training, we wrote an extensively detailed coding scheme describing exactly how to code each look that was handed to our coders. Throughout the initial months of training, we meet with the coders on a weekly basis to discuss questions and individual frames that looked ambiguous. After each session, we would revise the coding scheme to incorporate additional details, aiming to make the coding process progressively less subjective. During this period, every coder analysed the same interactions, and inter-rater reliability (IRR) was assessed weekly, comparing their evaluations with mine (Marta). With time, the coders had fewer questions and IRR increased. At that point, we deemed them sufficiently trained, and began assigning them different interactions from each other. Periodically, though, we all assessed the same interaction and meet to review and discuss our coding outputs.

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